

15.6 Using the NLS panel data on $N = 716$ young women, we consider only years 1987 and 1988. We are interested in the relationship between $\ln(WAGE)$ and experience, its square, and indicator variables for living in the south and union membership. Some estimation results are in Table 15.10.

TABLE 15.10 Estimation Results for Exercise 15.6

	(1) OLS 1987	(2) OLS 1988	(3) FE	(4) FE Robust	(5) RE
C	0.9348 (0.2010)	0.8993 (0.2407)	1.5468 (0.2522)	1.5468 (0.2688)	1.1497 (0.1597)
$EXPER$	0.1270 (0.0295)	0.1265 (0.0323)	0.0575 (0.0330)	0.0575 (0.0328)	0.0986 (0.0220)
$EXPER^2$	-0.0033 (0.0011)	-0.0031 (0.0011)	-0.0012 (0.0011)	-0.0012 (0.0011)	-0.0023 (0.0007)
$SOUTH$	-0.2128 (0.0338)	-0.2384 (0.0344)	-0.3261 (0.1258)	-0.3261 (0.2495)	-0.2326 (0.0317)
$UNION$	0.1445 (0.0382)	0.1102 (0.0387)	0.0822 (0.0312)	0.0822 (0.0367)	0.1027 (0.0245)
N	716	716	1432	1432	1432

(standard errors in parentheses)

- The OLS estimates of the $\ln(WAGE)$ model for each of the years 1987 and 1988 are reported in columns (1) and (2). How do the results compare? For these individual year estimations, what are you assuming about the regression parameter values across individuals (heterogeneity)?
- The $\ln(WAGE)$ equation specified as a panel data regression model is

$$\ln(WAGE_{it}) = \beta_1 + \beta_2 EXPER_{it} + \beta_3 EXPER_{it}^2 + \beta_4 SOUTH_{it} + \beta_5 UNION_{it} + (u_i + e_{it}) \quad (XR15.6)$$

- Explain any differences in assumptions between this model and the models in part (a).
- Column (3) contains the estimated fixed effects model specified in part (b). Compare these estimates with the OLS estimates. Which coefficients, apart from the intercepts, show the most difference?
- The F -statistic for the null hypothesis that there are no individual differences, equation (15.20), is 11.68. What are the degrees of freedom of the F -distribution if the null hypothesis (15.19) is true? What is the 1% level of significance critical value for the test? What do you conclude about the null hypothesis?
- Column (4) contains the fixed effects estimates with cluster-robust standard errors. In the context of this sample, explain the different assumptions you are making when you estimate with and without cluster-robust standard errors. Compare the standard errors with those in column (3). Which ones are substantially different? Are the robust ones larger or smaller?
- Column (5) contains the random effects estimates. Which coefficients, apart from the intercepts, show the most difference from the fixed effects estimates? Use the Hausman test statistic (15.36) to test whether there are significant differences between the random effects estimates and the fixed effects estimates in column (3) (Why that one?). Based on the test results, is random effects estimation in this model appropriate?

- For column 1 and 2, all of the coefficients are almost the same. These OLS regressions treat each year as a separate cross section and assume that omitted individual effects do not bias the estimates. In addition, the observations are assumed to be independent across individuals.
- The error term of panel data model is decomposed into two parts: an individual-specific effect u_i and an idiosyncratic error e_{it} . The individual effect captures unobserved individual heterogeneity that does not change over time. In contrast, the pooled OLS model treats all observations as independent and ignores these fixed individual characteristics.
- The fixed effects (FE) estimates differ noticeably from the pooled OLS estimates, especially for $EXPER$. In the pooled OLS model, the coefficient on $EXPER$ is about 0.127, while in the FE model it falls to 0.0575. This suggests that pooled OLS overestimates the

return to experience because it does not control for unobserved individual heterogeneity. The coefficient on UNION also decreases under FE estimation, implying that part of the positive union effect in pooled OLS may actually reflect unobserved worker characteristics.

Overall, the differences between FE and pooled OLS indicate that unobserved individual effects are important and correlated with the explanatory variables.

d. Degree of freedom: $df_1 = N - 1 = 715$, $df_2 = NT - N - K = 1432 - 716 - 4 = 712$.

$F_{0.01}(715,712) = 1.2$, since $11.68 > 1.2$, we have enough evidence to reject H_0

Hence, we can conclude that significant individual fixed effects exist.

e. The conventional fixed effects standard errors in column (3) assume homoskedasticity and no serial correlation in the error terms. In contrast, the cluster-robust standard errors in column (4) allow for heteroskedasticity and within-individual correlation over time.

Column (4) reports the fixed effects estimation with cluster-robust standard errors. Compared with the conventional FE standard errors in column (3), several robust standard errors become larger, especially for SOUTH. This suggests the possible presence of heteroskedasticity or within-individual correlation in the error terms.

f. The random effects (RE) estimates in column (5) differ from the fixed effects (FE) estimates in column (3), especially for EXPER. The RE coefficient on EXPER is larger than the FE estimate, although the single-coefficient Hausman comparison does not provide sufficient evidence to reject the null hypothesis that the FE and RE estimates are equal.

$$(15.36) \text{ gives that } t = \left| \frac{b_{FE,k} - b_{RE,k}}{\left[se(b_{FE,k})^2 - se(b_{RE,k})^2 \right]^{0.5}} \right| = \left| \frac{0.0575 - 0.0986}{(0.0328^2 - 0.022^2)^{0.5}} \right| = |-1.69| < 1.96 =$$

$z_\alpha(0.05)$. Based on the single-coefficient Hausman comparison for EXPER, we do not reject the null hypothesis that the FE and RE estimates are equal. Therefore, there is no strong evidence that the individual effects are correlated with the explanatory variables.

$$(cov(x_{it}, u_i) = 0)$$

15.20 This exercise uses data from the STAR experiment introduced to illustrate fixed and random effects for grouped data. In the STAR experiment, children were randomly assigned within schools into three types of classes: small classes with 13–17 students, regular-sized classes with 22–25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools. Data for the kindergarten classes are contained in the data file *star*.

- Estimate a regression equation (with no fixed or random effects) where *READSCORE* is related to *SMALL*, *AIDE*, *TCHEXPER*, *BOY*, *WHITE_ASIAN*, and *FREELUNCH*. Discuss the results. Do students perform better in reading when they are in small classes? Does a teacher's aide improve scores? Do the students of more experienced teachers score higher on reading tests? Does the student's sex or race make a difference?
- Reestimate the model in part (a) with school fixed effects. Compare the results with those in part (a). Have any of your conclusions changed? [Hint: specify *SCHID* as the cross-section identifier and *ID* as the "time" identifier.]
- Test for the significance of the school fixed effects. Under what conditions would we expect the inclusion of significant fixed effects to have little influence on the coefficient estimates of the remaining variables?
- Reestimate the model in part (a) with school random effects. Compare the results with those from parts (a) and (b). Are there any variables in the equation that might be correlated with the school effects? Use the LM test for the presence of random effects.
- Using the *t*-test statistic in equation (15.36) and a 5% significance level, test whether there are any significant differences between the fixed effects and random effects estimates of the coefficients on *SMALL*, *AIDE*, *TCHEXPER*, *WHITE_ASIAN*, and *FREELUNCH*. What are the implications of the test outcomes? What happens if we apply the test to the fixed and random effects estimates of the coefficient on *BOY*?
- Create school-averages of the variables and carry out the Mundlak test for correlation between them and the unobserved heterogeneity.

(a) For OLS, we have the following regression: $\text{readscore} = 437.7643 + \text{small} * 5.822816 + \text{aide} * 0.8178369 + \text{tchexper} * 0.4924687 + \text{boy} * -6.156421 + \text{white_asian} * 3.905809 + \text{freelunch} * -14.77134$

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	437.76425	1.34622	325.180	< 2e-16	***
small	5.82282	0.98933	5.886	4.19e-09	***
aide	0.81784	0.95299	0.858	0.391	
tchexper	0.49247	0.06956	7.080	1.61e-12	***
boy	-6.15642	0.79613	-7.733	1.23e-14	***
white_asian	3.90581	0.95361	4.096	4.26e-05	***
freelunch	-14.77134	0.89025	-16.592	< 2e-16	***

Since the coefficient of the small is significantly positive, students perform better in reading tests when they're in small classes.

Since the coefficient of the aide is not statistically significant, students perform the same in reading tests when having a teacher aide.

Since the coefficient of the tchexper is significantly positive, students perform better in reading tests for more experienced teacher.

Since the coefficient of the boy is significantly negative, boys perform worse in reading

tests.

Since the coefficient of `white_asian` is significantly positive, white and Asian students tend to perform better in reading tests.

Since the coefficient of `freelunch` is significantly negative, students receiving free lunch tend to perform worse in reading tests.

(b) For school fixed effects, we have the regression $\text{readscore_it} = \alpha_i + 6.490231 * \text{small} + 0.9960875 * \text{aide} + 0.2855668 * \text{tchexper} + -5.455941 * \text{boy} + 8.028019 * \text{white_asian} + -14.59357 * \text{freelunch} + u_{it}$

Coefficients:

	Estimate	Std. Error	t-value	Pr(> t)	
<code>small</code>	6.490231	0.912962	7.1090	1.313e-12	***
<code>aide</code>	0.996087	0.881693	1.1297	0.2586	
<code>tchexper</code>	0.285567	0.070845	4.0309	5.629e-05	***
<code>boy</code>	-5.455941	0.727589	-7.4987	7.440e-14	***
<code>white_asian</code>	8.028019	1.535656	5.2277	1.777e-07	***
<code>freelunch</code>	-14.593572	0.880006	-16.5835	< 2.2e-16	***

The relationship remains the same for all variables, but the coefficient of `white_asian` becomes much larger after controlling for school heterogeneity.

(c) For the test of the significance of school fixed effects, we have

Table: Fixed effects test: Ho: 'No fixed effects'

df1	df2	statistic	p.value	method	alternative
78	5681	16.69783	0	F test for individual effects	significant effects

Since the p-value is essentially zero, we reject the null hypothesis of no fixed effects and conclude that school fixed effects are statistically significant.

Significant fixed effects may have little influence on the coefficient estimates if the explanatory variables are only weakly correlated with the unobserved school effects.

(d) For the random effects model, we have the regression $\text{readscore_it} = 436.1268 + 6.458722 * \text{small} + 0.992146 * \text{aide} + 0.3026787 * \text{tchexper} + -5.512081 * \text{boy} + 7.350477 * \text{white_asian} + -14.58433 * \text{freelunch} + u_{it}$

Table: Random effects results for the reading score equation

term	estimate	std.error	statistic	p.value
(Intercept)	436.1268	2.0648	211.2217	0.0000
small	6.4587	0.9125	7.0777	0.0000
aide	0.9921	0.8812	1.1260	0.2602
tchexper	0.3027	0.0703	4.3060	0.0000
boy	-5.5121	0.7276	-7.5753	0.0000
white_asian	7.3505	1.4314	5.1353	0.0000
freelunch	-14.5843	0.8747	-16.6740	0.0000

Table: LM test for random effects in the reading score equation

statistic	p.value	method	alternative
81.71546	0	Lagrange Multiplier Test - (Honda)	significant effects

The LM test strongly rejects the null hypothesis of no random effects because the p-value is essentially zero. Therefore, random effects are statistically significant, and the pooled OLS model is inappropriate.

Although the LM test only indicates the presence of random effects, variables such as white_asian, freelunch, and tchexper may plausibly be correlated with unobserved school quality because better schools may have different student compositions and teacher characteristics.

(e) Using t-test equation, we have

At the 5% significance level, the critical value is approximately ± 1.96 . The calculated t-statistics for SMALL, AIDE, TCHEXPER, WHITE_ASIAN, and FREELUNCH are 1.146, 0.128, -1.938, 1.218, and -0.096, respectively.

Since none of the absolute t-values exceed 1.96, we fail to reject the null hypothesis that the fixed effects and random effects estimates are equal for these coefficients.

Therefore, there is no strong evidence of significant differences between the fixed effects and random effects estimates for these variables.

With joint test, we have

Table: Hausman test comparing fixed and random effects models

statistic	p.value	parameter	method	alternative
13.80915	0.0318423	6	Hausman Test	one model is inconsistent

Since the p-value is 0.0318, we reject the null hypothesis at the 5% significance level. Therefore, some explanatory variables are likely to be correlated with the unobserved school effects, implying that the random effects estimator is inconsistent.

This result differs somewhat from the individual t-tests above. Although none of the individual t-tests are significant at the 5% level, the joint Hausman test rejects the random effects specification. This suggests that the coefficients may differ jointly even if the individual differences are not statistically significant.

For the variable BOY, the t-statistic is 6.617, which exceeds the 5% critical value of ± 1.96 . Therefore, we reject the null hypothesis that the fixed effects and random effects estimates are equal for BOY. The variable boy is largely time-invariant within individuals in standard panel settings, so applying the Hausman comparison to this type of variable may not be very informative. However, in this grouped-data setting, the variable still varies within schools and therefore remains estimable.

(f) The Mundlak test was implemented by augmenting the random effects model with the school-average variable mean_freelunch.

The null hypothesis is that the coefficient on mean_freelunch is zero, implying that freelunch is uncorrelated with the unobserved school effects.

Since the p-value is 0.5646, we fail to reject the null hypothesis. Therefore, there is no strong evidence that freelunch is correlated with the unobserved school effects.

This result differs from the Hausman test, which rejected the random effects assumption. One possible explanation is that the Mundlak specification only included mean_freelunch rather than the full set of school-average variables.